INDEPENDENT BUILD TASK / SCORE IMPROVEMENT

Make evidence retrieval more dependable

Own the document and structured-data ingestion layer so identity-linked evidence is parsed consistently, time-bounded correctly, and cited with its source provenance. This work is separate from the main agent's planner and state-linking changes.

Goal	Your branch
Improve evidence coverage and reduce missed or malformed records without guessing case membership or fitting hidden answers. Keep all evidence local.	`codex/person-b-evidence-reliability` Base: latest `origin/codex/integration`

Why this matters

The current reproducible public benchmark is strong on status (0.9062) but has lower step recall (0.8201) and sequence accuracy (0.2188). The supported-case run scores 29 public examples; three cases stop for manual review because the engine cannot link a resource. Full-dataset accuracy is not known because no full-set gold labels are provided. Your task is to find general ingestion gaps that could cause real evidence to be missed, then protect each fix with tests.

Scope and boundaries

- Own: `src/ghostdata/loaders/`, loader-focused tests, and a concise findings note at `docs/evidence_loader_audit.md`.
- Coordinate before changing shared contracts. Avoid identity resolution, state reconstruction, evidence-pipeline joins, policy, planner, API, and UI files; another workstream owns those.
- Use participant-facing files under `dataset/` only. Never read or infer from organizer-only data.
- Do not add case IDs, file IDs, expected answers, or benchmark-family shortcuts to production parsing code.
- Do not copy dataset files into commits or tests. Use small fictional fixtures created by the tests.

Work plan

STEP	WHAT TO DO
01 Establish a baseline	Fetch first. Reproduce the current supported-public-sample metrics using the existing evaluator. Record the exact command and the three safe-review cases; do not edit the gold sample.
02 Audit parsers by source	Trace the manifest through PDF/image extraction, JSON/record parsing, log parsing, and SQL snapshots. Check timestamps, evidence IDs, replica labels, unreadable-file warnings, and cache behavior.
03 Fix only demonstrated gaps	Make narrow, format-agnostic loader fixes. Preserve source provenance and capture/observation times. Keep operational results marked as claims; parsing must not turn a workflow claim into a verified record state.
04 Add regression coverage	Add at least 8 focused tests using fictional fixtures across the formats below. Include corrupt or unreadable evidence and cutoff boundaries. Keep optional OCR tests deterministic when Tesseract is missing.

IMPLEMENTATION CHECKLIST

Cover the formats that hide real records

Write small fixtures that test parser behavior and provenance, not exact challenge-case identifiers. Keep extraction separate from policy decisions.

SOURCE	CHECK	PASS CONDITION
PDF / image	Embedded text, scanned pages, OCR absent or failing	Text and confidence are correct; unreadable sources return a clear warning and are not silently treated as empty evidence.
JSON	Nested records, inherited system/time, null and direct identifier fields	Resource IDs, category, source ID, and observed timestamp survive parsing.
Logs	App traces, syslog, nested messages, malformed lines	Resource links are parsed without mistaking event/service names for resources; activity remains claim-only.
SQL	Primary and replica snapshots, timestamps, escaped JSON values	Rows preserve system/replica identity and temporal metadata; supported SQL dialects fail safely when malformed.
Manifest / cache	Path traversal, missing files, cache hit and corrupt cache	Paths stay under the dataset root; cache errors do not change source content or evidence identity.

Required checks before handing back

- Run `python -m pytest -q`, `ruff check src/ghostdata/loaders tests`, and `ruff format --check src/ghostdata/loaders tests`.
- Re-run the supported public-sample evaluation. Report status accuracy, step precision/recall/F1, sequence accuracy, and any changed cases.
- If metrics change, explain why each change is supported by participant-facing evidence or a general parser defect. Do not optimize a metric by hardcoding public sample outputs.
- Keep the existing full `output.csv` and teammate files untouched. Do not merge directly into integration; return a commit for review.

Done means

Code Narrow loader fixes only, with at least 8 new fictional regression tests.	**Evidence** A short audit note describing the defect, source type, and why the fix generalizes.	**Review** Commit SHA, commands run, before/after public metrics, and any remaining OCR limitations.

Do not promise a score increase before the evidence-backed changes are measured. The objective is reliable extraction that improves recall without adding unsupported findings.